How Ultrasound Can Capture Capacitor Defects

Most MLCCs are relatively inexpensive, but when a capacitor fails in service, the consequences may be anything from a slight change in a video image to the complete failure of a system. So their testing is important

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Multi-layer ceramic chip capacitors (MLCCs) work by storing energy and releasing it when it is needed. They consist of stacks having alternating layers of a thin metal electrode and a thicker ceramic dielectric, or insulator. Because of their layered structure, MLCCs are ideal for inspection by an acoustic micro imaging (AMI) tool.

This tool sends a pulse of ultrasound into the surface of the MLCC and receives and analyses the return echoes from within the part. It performs this task up to 30,000 times per second while scanning across the MLCC just above its surface. The end result is an acoustic image of the MLCC's interior, including structural defects.

The echoes come only from material interfaces—where a material is in contact with another material. In a semiconductor device, this might be the interface between mould compound and silicon, which would give a strong echo because the acoustic impedance (acoustic velocity x density) values of the two materials are so different.

The operator of an AMI tool examining an IC device may be dealing with features at a few to several depths, but even small low-voltage MLCCs may have hundreds of alternating layers of ceramic and metal. At interfaces between solid materials, part of the pulse is reflected and part transmitted through the interface. If the acoustic impedances of ceramic and metal were very different from each other, a launched pulse could be completely attenuated after the first few layers. But the acoustic impedances of the ceramic and metal in MLCCs are slightly different, so echoes can be pulsed into and reflected from the bottom layer of the MLCC, even if there are hundreds of layers.

Electrical failure of an MLCC may begin with an air inclusion trapped during fabrication of the component. The inclusions can broadly be separated into two types: voids

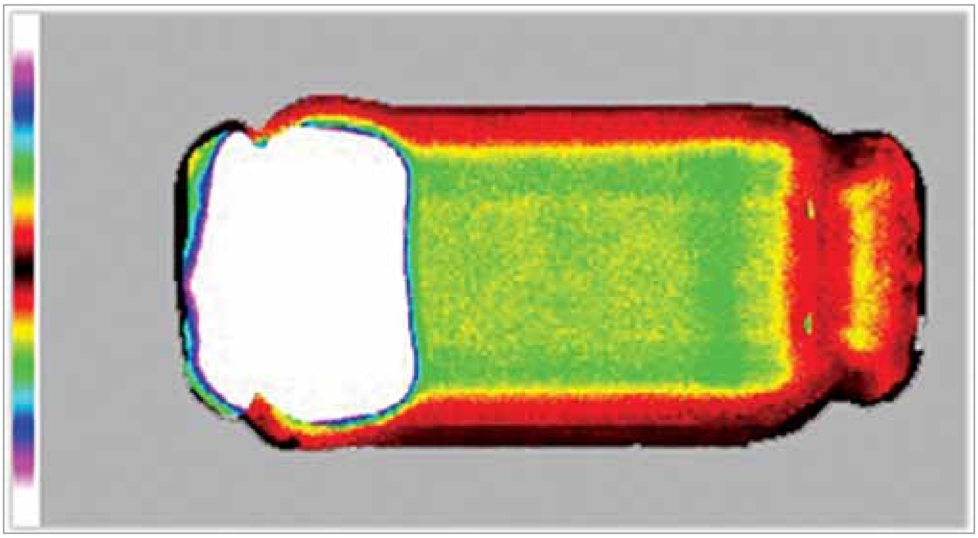

Fig. 1: Acoustic image of an MLCC having a large internal delamination (white)